

January 28, 2026

ECEN-607: Advanced Analog IC Design

Assignment #2

Due: 02/05/2026

Homework will not be received after due.

Upload your solutions through Canvas

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Design at transistor level a two-stage amplifier; use the following specifications:

$$V_{DD} = 1.8 \text{ V}$$

$$V_{SS} = 0 \text{ V}$$

$$\text{DC Gain} > 40 \text{ dB}$$

Employ miller compensation to stabilize the amplifier; loop phase margin $> 45^\circ$. Your amplifier must also satisfy the following specifications:

$$\text{GBW} > 4 \text{ MHz}$$

Load Capacitor = 100-500 pF (amplifier's phase margin must be maintained over 45 degrees for the entire range of CL).

V_{DSAT} for the transistors, should be in the range $\{150\text{mV} - 250\text{mV}\}$.

Current sources needed for biasing the amplifier stages can be implemented employing ideal current sources.

Results:

- i) Show amplifier's functionality. Simulate the amplifier and include in your report screen shots of both magnitude and phase response. Report the values of the unity gain frequency, phase margin and gain margin.
- ii) Implement a capacitive amplifier with an inverting gain of 6dB, $\beta=1/3$; minimum capacitor used in the capacitive amplifier must be $\geq 100\text{fF}$. You may need to connect a large resistor ($>1\text{Mohm}$) in parallel with the feedback capacitor; why?
 - a. Simulate the amplifier and report both magnitude and phase response.
 - b. Plot the step response for an input step of 0 to 100mV variation. Report the 1% settling time and value of the settling error. Ideal output voltage variation should be -200mV.
 - c. Comment on the results.

- iii) Repeat ii) for the case of an inverting amplifier with gain=12dB; evidently the feedback factor is different now. For the transient simulation, reduce the size of the input step to 0-50mV.

Please discuss your results; include necessary screen shots.